METERIALHUB

On-Demand Hyperlocal Construction Materials Delivery Platform

Connecting buyers with nearby vendors for fast, transparent, location-based material delivery.

PROJECT ABSTRACT

Construction and renovation projects frequently experience delays due to sudden shortages of essential materials such as cement, steel rods (kambi), bricks, and sand. MeterialHub is a hyperlocal, on-demand construction materials marketplace that connects buyers with nearby vendors and enables fast, location-based delivery to construction sites — reducing procurement time from days to hours while improving transparency and operational efficiency.

1 PROBLEM STATEMENT

- Sudden shortages of construction materials can interrupt work and increase project costs.
- Direct procurement from large industries is often slow and oriented toward bulk purchasing.
- There is no centralized location-based system for discovering nearby construction material vendors.
- Buyers have limited real-time visibility into vendor stock, order status, and delivery progress.
- Manual coordination between vendors and drivers can delay material delivery to construction sites.

2 PROPOSED SOLUTION

MeterialHub is a multi-role web platform connecting Buyers, Vendors/Companies, Delivery Partners, and Administrators in a single construction-material procurement and delivery ecosystem. Location-based services help identify nearby vendors, while the platform coordinates ordering, inventory, delivery assignment, payments, and real-time status updates.

3 USER ROLES & DOMAIN STRUCTURE

Buyer

`meterialhub.com`

Browse nearby vendors, place orders, select payment mode, track delivery, view order history, and rate vendors/drivers.

Vendor / Company

`vendor.meterialhub.com`

Manage products, inventory, pricing, orders, driver assignment, and earnings.

Delivery Partner

`driver.meterialhub.com`

Manage assigned deliveries, pickup/drop-off, delivery status, COD collection, and trip history.

Admin

`admin.meterialhub.com`

Verify vendors/drivers, monitor orders and payments, manage disputes, commissions, and analytics.

4 CORE MODULES

1. **Location-Based Vendor Discovery**
   Detect buyer location and display nearby vendors within a configurable radius; calculate distance and ETA.

2. **Product & Inventory Management**
   Manage construction materials, price, unit, and live stock; hide unavailable items from buyer search.

3. **Order Management**
   Process orders through Placed → Confirmed → Assigned to Driver → Out for Delivery → Delivered.

4. **Delivery Assignment & Tracking**
   Assign available drivers and provide pickup/drop details with real-time delivery progress.

5. **Payment Management**
   Support COD with cash collection recording and an online payment option; gateway to be finalized.

6. **Administration & Analytics**
   Manage verification, orders, payments, disputes, commissions, reporting, and platform operations.

5 TECHNOLOGY STACK

FRONTEND **Next.js + TypeScript**	STYLING **Tailwind CSS**	BACKEND **Python FastAPI**	DATABASE **PostgreSQL**
AUTHENTICATION **JWT Role-Based Auth**	LOCATION **Map / Geolocation APIs**	PAYMENTS **COD + Online**	REAL-TIME **WebSockets**

6 KEY BENEFITS

✓ Faster procurement for urgent and small-quantity construction material requirements.

✓ Transparent and trackable delivery operations.

✓ Centralized operational oversight, dispute management, commissions, and reporting.

✓ Greater visibility and business opportunities for local vendors.

✓ Flexible payment options for buyers.

CONCLUSION

MeterialHub provides a digital, hyperlocal approach to construction material procurement by connecting buyers with nearby vendors and coordinating last-mile delivery through Delivery Partners. By integrating product and inventory management, location-based discovery, order processing, delivery tracking, and payment management under centralized administration, the platform aims to make construction material sourcing faster, more transparent, and operationally efficient.